Prabhavit Gupta

AI Data Enablement Engineer | Semantic Layers, Governed Data Products, RAG & PySpark/dbt

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Executive Summary

- **AI Data Enablement & Semantic Foundations:** Data Enablement and Applied AI Engineer with 3 years commercial software experience engineering trusted data foundations, semantic models, and natural-language query interfaces across European enterprise systems.
- **Governed Data Products & Production RAG:** Specialized in architecting governed data pipelines (PySpark, dbt, SQL, Airflow) and conversational analytics interfaces (Databricks Genie spaces, Streamlit) grounded in structured enterprise data with strict RBAC, row/column-level security, and auditability.
- **Production Optimization & Business Alignment:** Deep expertise bridging technical data platforms and business stakeholders; optimizing warehouse compute, semantic caching, token consumption, and model routing to ensure deterministic, cost-effective natural language data exploration.

Technical Skills & Core Competencies

- **Data Enablement & Semantic Layers:** Semantic Data Modeling, Business Glossaries, KPI Hierarchies, Databricks Genie Spaces, Unity Catalog Metadata Management, Governed Data Products.
- **Data Engineering & Pipelines:** PySpark, dbt, SQL, Apache Airflow, Python (AsyncIO, Pydantic), PostgreSQL, Vector Stores, ETL/ELT Orchestration, Data Lineage & Auditing.
- **Conversational Analytics & RAG:** Enterprise RAG Architectures, Streamlit/FastAPI NL-to-SQL Interfaces, Context Management, Guardrails, Evaluation Baselines (RAGAS, DeepEval).
- **Data Governance & Security:** Role-Based Access Control (RBAC), Row/Column-Level Security, PII Masking, Regulated Environment Compliance, Telemetry & Observability (Langfuse).
- **Cloud & Platforms:** Databricks, AWS, Azure, Docker, Git, CI/CD GitHub Actions, Query Performance Optimization, Compute Sizing & Token Budgeting.

Professional Experience

AI Data Enablement & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Italy)

- Partnered with European enterprise client teams to build and deploy AI-enablement layers bridging complex enterprise data lakes and business user natural-language analytics.
- Designed semantic models and governed data products, aligning technical schemas with business glossaries and KPI definitions to enable non-technical teams to query data accurately without manual SQL.
- Built and evaluated production RAG pipelines and conversational data applications using Streamlit and FastAPI, incorporating deterministic validation, semantic caching, and strict data grounding.
- Implemented rigorous enterprise data governance protocols including role-based access control (RBAC), row/column-level filtering, and audit logging to ensure compliant data access across multi-tenant environments.
- Profiled and optimized end-to-end platform costs and performance, tuning data pipeline compute clusters, caching query embeddings, and reducing redundant LLM inference overhead by 45%.
- Established automated evaluation harnesses using DeepEval and RAGAS, benchmarking retrieval accuracy, semantic fidelity, and hallucination prevention prior to deployment.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed robust ETL/ELT pipelines and microservices in Python (FastAPI, PySpark) for automated document ingestion, structured schema validation, and database synchronization.
- Engineered high-performance hybrid retrieval layers combining PostgreSQL, pgvector, and Redis semantic caching, supporting reliable real-time search across hundreds of thousands of corporate records.
- Designed modular REST APIs and automated CI/CD workflows (GitHub Actions), enforcing automated schema migrations, data consistency checks, and test coverage using PyTest.
- Collaborated with domain experts in finance and operations to map business requirements into resilient data architectures and automated reporting workflows.

Key Engineering Projects

- **Enterprise Conversational Data Engine (Databricks + PySpark + Streamlit + RAG):** Built a natural-language data interrogation platform enabling cross-functional stakeholders to extract governed metrics and insights, enforcing schema consistency and automated query validation.

- **Governed Semantic Layer & Retrieval Pipeline (dbt + PostgreSQL + FastAPI + pgvector):** Engineered a metadata-driven semantic layer mapping domain entities to vector embeddings and structured SQL tables, complete with automated RBAC security guards and Langfuse tracing.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Focus on formal logic, decision theory, economic modeling, and quantitative analytical methods.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Indefinite right to work in Italy and the EU; Italian Partita IVA ready for immediate B2B contractor invoicing.